Iulian Oleniuc

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Education

Master's Degree — 2023–2025

Institution: Faculty of Computer Science Iași
Program: Advanced Studies in Computer Science
Thesis: In my MSc thesis, "A Hybrid Key-Policy Attribute-Based Encryption Scheme for Universal Circuits, Based on Monotone Span Programs," building on top of the Hu–Gao and Goyal et al. schemes, I developed the **pairing-based KP-ABE** scheme with, I argue, **the highest degree of generalization**. A paper on part of this work was accepted at the **CSJM 2025** journal.
Thesis grade: 9.5

Bachelor's Degree — 2020–2023

Institution: Faculty of Computer Science Iași
Thesis: In my BSc thesis, "Heuristic Optimizations for Boolean Formulas with Applications in Attribute-Based Encryption," I developed four heuristics to **optimize monotone Boolean formulas**. These optimizations enhanced the performance of a **circuit-based KP-ABE** scheme by up to 60% in real-world scenarios. A paper on this work was accepted at the **KES 2023** conference.
Thesis grade: 10

High School — 2016–2020

Institution: Colegiul Național "Emil Racoviță" Iași
Specialization: Mathematics–Informatics

Experience

Status 32 — Oct 2025–Present

I am developing full-stack interfaces for a **CRM application**, using React, **DevExtreme**, .NET, and SQL. I have learned about the **CQRS design pattern**.

Contests

Algorithms — High School–College

National Olympiad in Informatics: Participation (9th grade), Bronze (10th grade), Silver (11th grade), Qualification (12th grade)
ICPC Regional Contest Iași: 1st place (3rd year of BSc)

Languages

- Romanian — Native
- English — Advanced

Skills

- Problem solving
- Teaching
- Technical writing
- Teamwork
- Critical thinking

Knowledge

- Algorithms
- Data structures
- Cryptography
- Machine learning
- Neural networks
- Web developing
- Relational databases
- Object-oriented programming

Technologies

- C++, Java, Python
- HTML, CSS, JavaScript
- React, Vue
- Gatsby, Nuxt, Astro
- FireBase, SupaBase
- PostgreSQL

Passions

- Cycling
- Books
- Movies
- Hip-hop

Projects

InfoGenius — Computer Science Blog 2017–Present

I founded InfoGenius in the 9th grade and have since written posts on topics such as famous algorithms, data structures, C++ programming, and National Olympiad problems, among others. With **over 300 comments**, the website has made a meaningful impact in **helping people** learn computer science, particularly in competitive programming.

InfoGym — Teaching Computer Science 2022–2025

InfoGym is a programming circle founded by eMAG as part of the **“Hai la Olimpiadă!”** program, where I have delivered several lectures to middle and high school students. This involved creating instructional papers, including **“Introduction to Combinatorics”** and **“Trie & the Knuth–Morris–Pratt Algorithm,”** which are among my proudest works.

FIICode — Organizing a Programming Contest 2021–2024

FIICode is a programming contest organized by students of the Faculty of Computer Science Iași. I contributed to the **organizing and scientific committees** for the 2021–2024 editions, focusing on algorithms. My responsibilities included designing problem ideas, generating test data, and preparing problem statements and editorials.

ICPC Training Camps — Being a Coach 2024

I participated in two **ICPC training camps** in 2024, coaching teams from the Faculty of Computer Science Iași. The camps took place in Osijek, Croatia, and Wrocław, Poland, where I also served on the scientific committee at the latter. Additionally, I coached at SEERC 2024 in Bucharest.

CruceaRosieIasi.ro — Website for the Iași Red Cross 2022

I developed a **fast, user-friendly website** for the Iași Red Cross using modern web technologies like Vue, Nuxt, and Firebase. The site includes an admin panel with a minimalist yet powerful **content management system**.

Publications

LSSS Limitations Over Boolean Circuits 2025

A paper co-authored with Alex Ioniță, published in CSJM, 33(1), 2025. It offers a gentle introduction into the realm of monotone span programs, linear secret sharing schemes, and attribute-based encryption.

Boolean Circuit Optimizations for ABE 2023

A paper co-authored with Alex Ioniță and Denis Banu, published at the KES 2023 conference. It introduces four heuristics for optimizing Boolean circuits in key-policy attribute-based encryption schemes.